Silicon NPN Power Transistors 2SC2553

DESCRIPTION

·With TO-220C package
·High collector breakdown voltage
: V_{CEO}=400V(Min)
·Excellent switching time
: t_r=1.0μs(Max.)
: t_f=1.0μs(Max.@I_C=4A

APPLICATIONS

·Switching regulator and high voltage switching applications
·High speed DC-DC converter applications

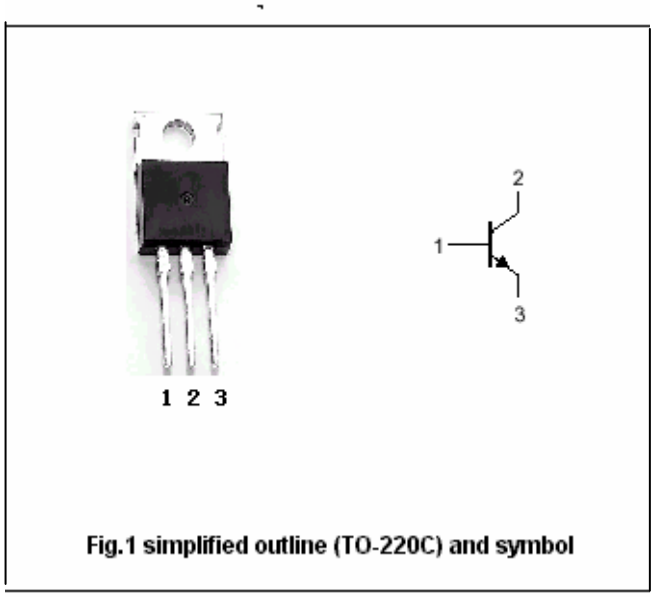


Fig.1 simplified outline (TO-220C) and symbol

PINNING

PIN	DESCRIPTION
1	Base
2	Collector;connected to mounting base
3	Emitter

Absolute maximum ratings(Ta=25□)

SYMBOL	PARAMETER	CONDITIONS	VALUE	UNIT
V_{CBO}	Collector-base voltage	Open emitter	500	V
V_{CEO}	Collector-emitter voltage	Open base	400	V
V_{EBO}	Emitter-base voltage	Open collector	7	V
I_C	Collector current		5	A
I_B	Base current		1	A
P_C	Collector dissipation	T_a=25□	1.5	W
		T_C=25□	40	
T_j	Junction temperature		150	□
T_{stg}	Storage temperature		-55~150	□

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CHARACTERISTICS

Tj=25□ unless otherwise specified

SYMBOL	PARAMETER	CONDITIONS	MIN	TYP.	MAX	UNIT
$V_{(BR)CEO}$	Collector-emitter breakdown voltage	I_C=10mA ; I_B=0	400			V
$V_{(BR)CBO}$	Collector-base breakdown voltage	I_C=1mA ; I_E=0	500			V
V_{CEsat}	Collector-emitter saturation voltage	I_C=5A; I_B=1A			1.0	V
V_{BEsat}	Base-emitter saturation voltage	I_C=5A; I_B=1A			1.5	V
I_{CBO}	Collector cut-off current	V_{CB}=400V ;I_E=0			100	μA
I_{EBO}	Emitter cut-off current	V_{EB}=7V; I_C=0			1	mA
h_{FE-1}	DC current gain	I_C=3A ; V_{CE}=5V	12			
h_{FE-2}	DC current gain	I_C=5A ; V_{CE}=5V	8			
Switching times						
t_r	Rise time	V_{CC}≈200V; I_C=4A I_{B1}=-I_{B2}=0.4A;R_L=50Ω Duty cycle≤1%			1.0	μs
t_{stg}	Storage time				2.5	μs
t_f	Fall time				1.0	μs

PACKAGE OUTLINE

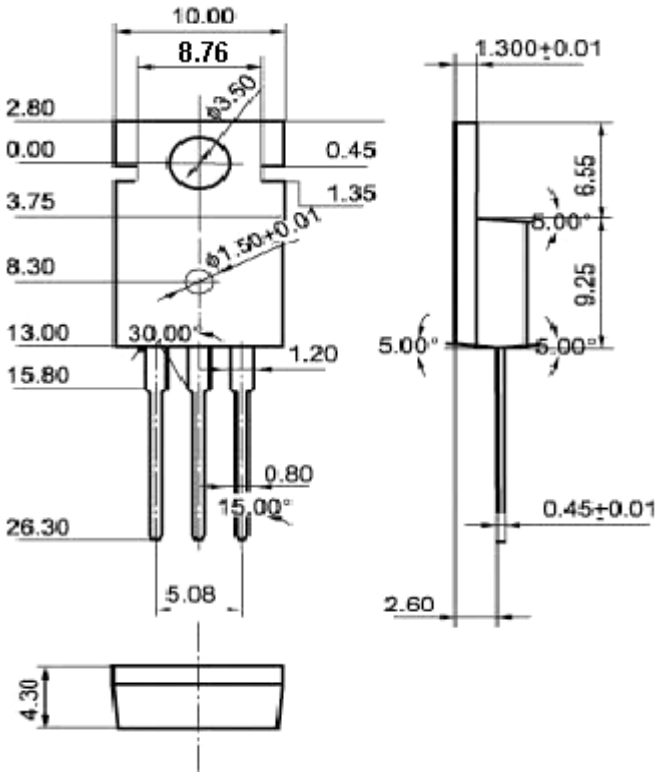


Fig.2 Outline dimensions (unindicated tolerance:±0.10 mm)

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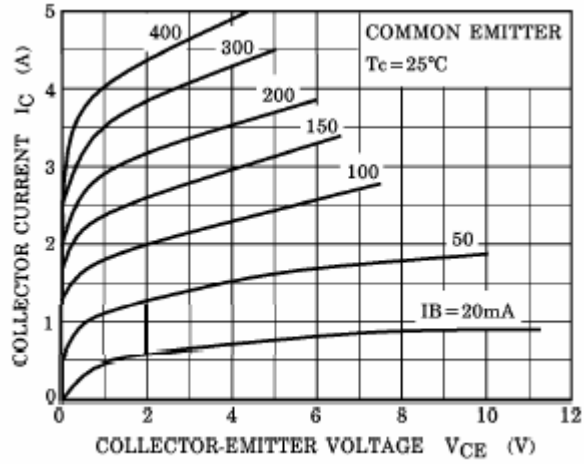


Fig.3 Static Characteristic

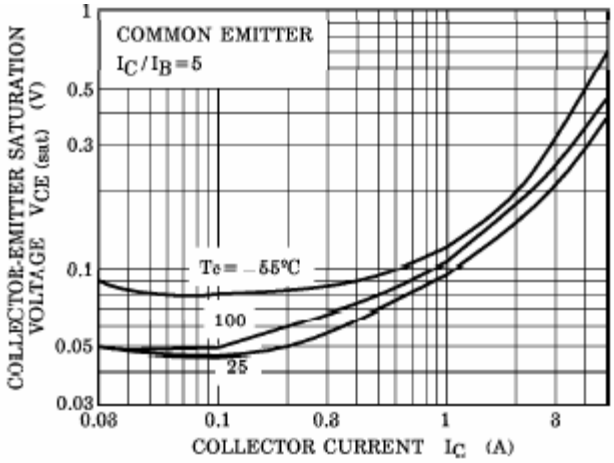


Fig.4 Collector-Emitter Saturation Voltage

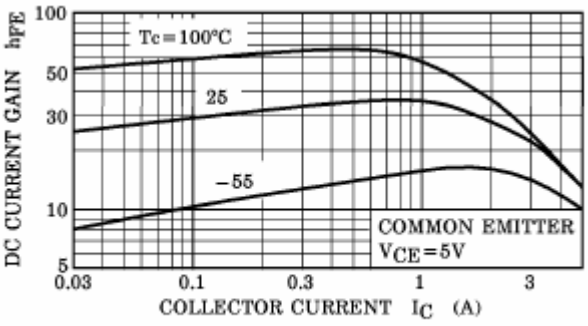


Fig.5 DC current Gain

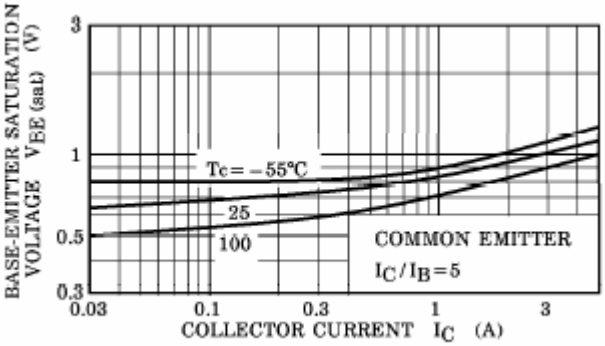


Fig.6 Base-Emitter Saturation Voltage

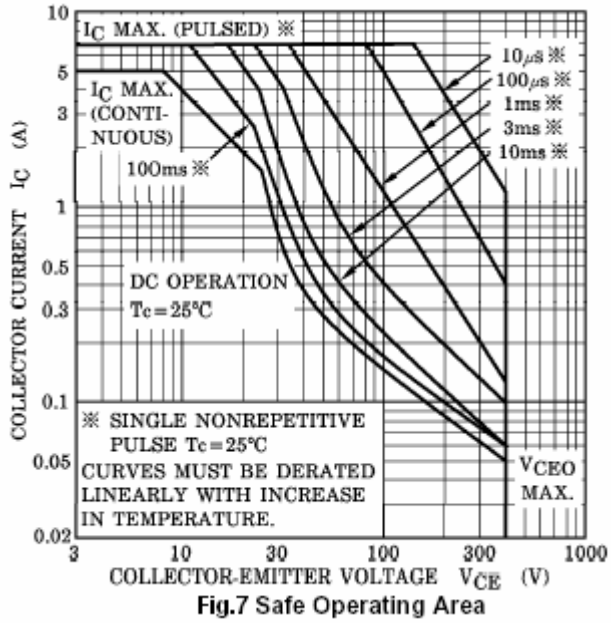


Fig.7 Safe Operating Area